Do Liquidated Wallets Speak a Different Language?

Reading wallet histories on a blockchain lending protocol (Aave v3, Ethereum) as sentences, and testing whether a language model of safe behavior can spot trouble

Muhammad Saad Maqsood · M.Sc. Natural Language Processing · Universität Trier · NLP module 2026

Code and reproduction:
github.com/MSaadMaqsood/wallet-behavior-as-language

● safe ● liquidated

1 Hypothesis

> **If a wallet's history is read as a sentence, do wallets that later get liquidated speak differently from safe ones, and can a language model hear it without prices or the health factor?**

Aave v3 is a lending protocol on the Ethereum blockchain: wallets deposit crypto as collateral, borrow against it, and every action is public. If the collateral loses too much value, the loan is liquidated. Liquidation studies start from prices and the health factor (Perez et al., 2021; Qin et al., 2021); DeepLog (Du et al., 2017) instead flagged system logs that surprised a language model of normal logs. I apply that idea to wallets.

H1 Liquidated wallets get a higher perplexity than safe ones.

H2 Perplexity alone beats chance (ROC-AUC).

H3 Action order carries information beyond counts.

H4 The signal survives strict leakage controls.

2 Methodology

Most common liquidated sentence (12.6 % of them):
`<BOS>` `SUPPLY_VOL` `BORROW_STABLE` `<EOS>`

Most common safe sentence (6.3 % of them):
`<BOS>` `SUPPLY_VOL` `BORROW_STABLE` `REPAY_STABLE` `WITHDRAW_VOL` `<EOS>`

Data. 4.1 M Aave v3 lending events from the Ethereum blockchain (Jan 2023 – Jul 2026) in block and log order; withdrawals through Aave's helper contracts go back to the real wallet.

Sentences. 10,056 liquidated and 71,285 safe wallets (borrowed, never liquidated); one token per action × asset class, only actions before the first liquidation.

Models. A Kneser–Ney 6-gram model written from scratch and a 2-layer Transformer (88k parameters), trained on safe wallets only; logistic regression as a baseline.

Matched windows. A liquidated history stops because of the liquidation, so every wallet is also cut to 4 actions ending mid-stream.

3 Results · test set: 14,257 safe and 2,012 liquidated wallets

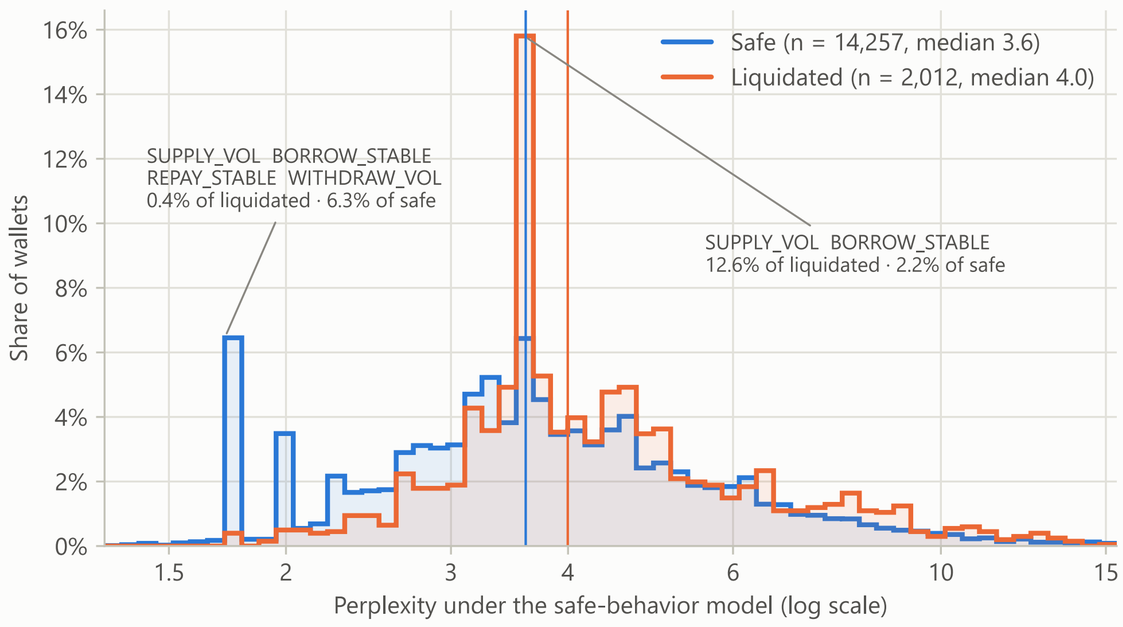


Fig. 1 Liquidated wallets surprise the safe model more (median 4.00 vs 3.60, p = 1.5×10^{-53}).

Table 1 ROC-AUC (0.5 = chance)

Score	Full history	4-action window
Perplexity, n-gram	**0.61**	**0.51**
end token only	0.79	–
actions only	0.45	–
Perplexity, Transformer	0.64	0.50
Bag-of-actions model	0.39	0.40
Classifier, action counts	0.81	0.63
Classifier, 1–3 action sequences	**0.85**	**0.73**

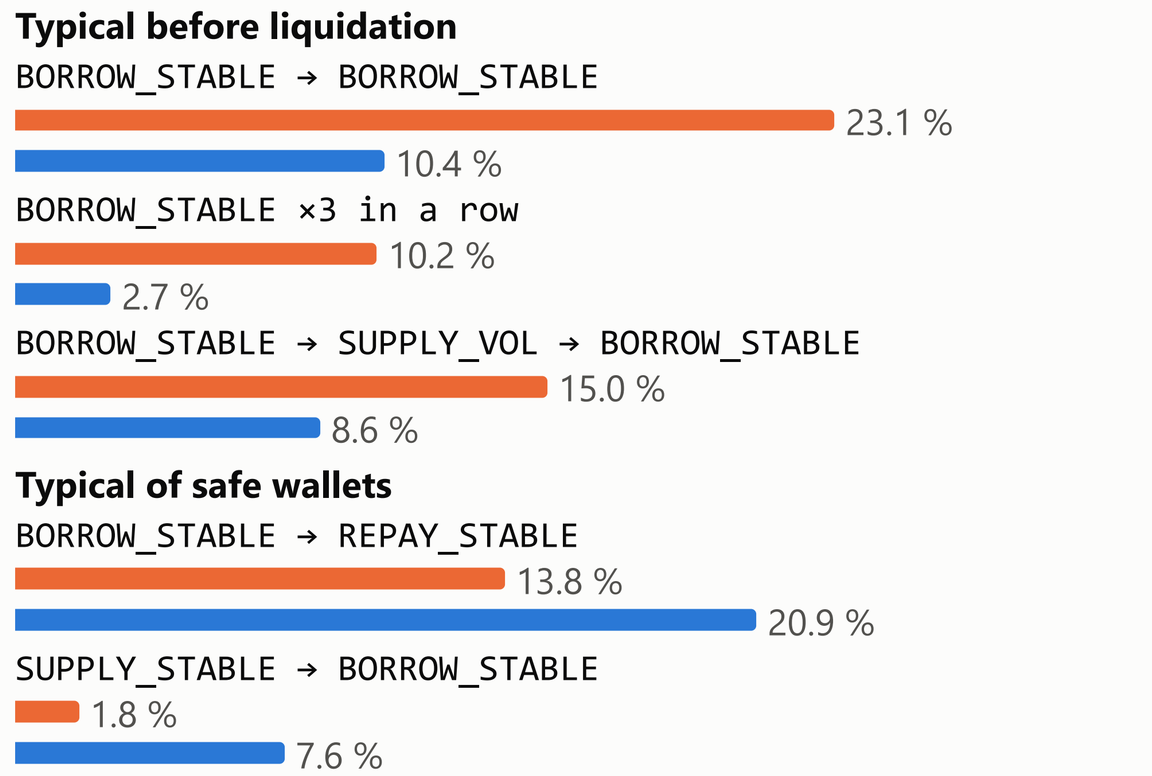


Fig. 2 Share of 4-action windows that contain a pattern

Ranked by weighted log-odds z-scores; all |z| > 11. Two borrows in a row: odds ratio 2.39.

What this means

H1 Holds, but only on full histories.

H2 Does not hold up: the separation sits in the end token. On matched windows both language models fall to chance, although the Transformer models safe behavior better (perplexity 3.54 vs 3.85).

H3 Holds: word order adds 0.096 AUC over counts (95 % CI 0.081–0.111).

H4 No leakage, but the cutoff created an artifact; the matched windows remove it.

> **Liquidated wallets have their own grammar, but they are not more surprising to a model of safe behavior: perplexity mostly caught unfinished sentences.**

Unexpected: counting never-borrowing wallets as safe, as an earlier non-NLP study did, lifts perplexity to AUC 0.66. **Limitations:** no prices by design · 8 tokens · recent safe wallets may still be liquidated · one protocol, one chain.

References (full citations in the appendix): Qin et al. 2021 · Perez et al. 2021 · Du et al. 2017. The idea for this system came to me while working on a blockchain project; the event pipeline is adapted from an earlier group project.